📡 High-Level Technical Modeling

Week 5 • Day 4 • Technical Infrastructure & Modeling

TPM Academy

🎯 Day 4 Objectives

- Draw a **happy-path sequence** with lifelines, protocols, latencies
- Draw a **sad-path sequence** showing user-visible recovery
- Draw a **weird-path sequence** with named invariant
- Verify total latency annotation matches **Technical Concept Document (TCD) Section 4 Service Level Objective (SLO) budget**
- Cross-review with another triad and use AI to surface gaps

🕑 Today's Shape

Clock	Block	Outcome
09:00 – 09:15	Opening	Pin TMD Sections 1–3; whiteboards out
09:15 – 10:30	Block 1 + Activity 1	Happy-path sequence
10:45 – 12:00	Block 2 + Activity 2	Sad-path sequence
13:00 – 14:15	Block 3 + Activity 3	Weird-path sequence
14:30 – 16:00	Block 4 + Activity 4 + Wrap	Cross-review + AI critique

Block 1 — The Happy Path

Time top to bottom; actors as lifelines

📝 Sequence Diagram Anatomy

Element	What it is
Lifeline (vertical)	An actor: user, mobile app, service, datastore, queue
Solid arrow	Synchronous call
Dashed arrow	Async / fire-and-forget
Return arrow (dotted)	Response
Note	Latency budget, condition, annotation
Alt / opt block	Conditional branch

5–12 lifelines. 10–25 arrows. More becomes unreadable.

🎯 Right Diagram for the Right Job

✅ Sequence is right for

- One user action's journey
- Service interactions during a transaction
- Order of events; where things go wrong

❌ Sequence is wrong for

- All possible interactions (use Container)
- Data structure (use an entity-relationship / ER diagram)
- Deployment topology (use cloud diagram)

📚 Three Sequences in TMD Section 4

1. **Happy path** — central success case (full detail)
2. **Sad path** — most common user/system error (abbreviated)
3. **Weird path** — edge case from your acceptance criteria (AC) "weird" coverage (abbreviated)

Sad and weird paths usually **diverge from the happy path at one arrow**. Draw from the divergence.

👥 Activity 1 — Happy Path

Triads • 35 minutes

- 5 min: identify actors (lifelines)
- 5 min: order them caller → called
- 15 min: draw messages with protocol labels
- 5 min: latency annotations per arrow
- 5 min: 60-sec walk to another triad

⏱️ 5 + 5 + 15 + 5 + 5

Block 2 — Sad Path

Where the user goes wrong; how we recover gracefully

📝 Sad Path Discipline

- Pick from Product Requirements Document (PRD) Section 6 sad-path AC
- Show the **divergence point** — the arrow where it goes differently
- Show the **status code + error body** from Section 3
- Show the **user's recovery action**

> **The user must have a path forward.** Not a dead-end.

👥 Activity 2 — Sad Path

Triads • 40 minutes

- 5 min: pick the sad path
- 20 min: draw the divergence
- 5 min: status code + error body
- 5 min: user recovery action
- 5 min: 30-sec walk

⏱️ 5 + 20 + 5 + 5 + 5

Block 3 — Weird Path

Network drop, race, timeout, boundary — with named invariant

📝 Weird Path Discipline

- Pick from PRD Section 6 weird-path AC
- Show **retry / idempotency / fallback** mechanics
- **Name the invariant** — the promise we keep even here

The named invariant is the senior-author signal. "At most one ReconcileEvent per Idempotency-Key per 24h."

📝 Worked Weird Path: Network Drop

```
Dispatcher     Mobile      API GW     Reconcile     Postgres
|              |           |          |             |
| tap submit   |           |          |             |
|────────────▶| POST ...  |          |             |
|              |  Idempotency-Key: K1|          |            |
|              |──────────▶|          |             |
|              |           |──────────▶             |
|              |           |          | INSERT (commit)
|              |           |          |───────────▶
|              |           |          | ◀── ok
|              |           | ◀── 201
 |
|              |  ✕ network drop
     |
| ◀───────────  user sees timeout
   |
| tap retry    |
                        |
|────────────▶| POST ... Idempotency-Key: K1 (same)|
|              |──────────▶| idempotency check     |
|              |           | → return cached 201   |
|              | ◀── 201
               |
| ◀───────────  modal closes; no duplicate        |
```

```
Invariant: at most one ReconcileEvent per (Idempotency-Key) per 24h
```

👥 Activity 3 — Weird Path

Triads • 40 minutes

- 5 min: pick the weird path
- 25 min: draw divergence + recovery
- 5 min: name the invariant
- 5 min: walk to another triad

⏱️ 5 + 25 + 5 + 5

Block 4 — Cross-Review + AI Critique

What did we forget?

🔎 Cross-Review Questions

1. **Happy path:** does latency add up? Protocols consistent with Section 3?
2. **Sad path:** does the user have a clear recovery action?
3. **Weird path:** is an invariant named? Is the recovery actually safe?

🤖 AI Critique Prompt

```
Role: Engineering reviewer with strong sequence-diagram instincts.
Context: <plain-text descriptions="" of="" the="" three="" sequence="" diagrams,="" including="" all="" lifelines,="" arrows,="" pro
Task: Identify what's missing, ranked by importance:
1. What sad/weird paths are NOT yet diagrammed but should be?
2. What arrow is missing from the happy path?
3. What invariant should be added to the weird path that isn't?
Constraints:
- Do not invent technologies
- Be specific
Format: Three sections; ranked items in each.</plain-text>
```

👥 Activity 4 — Cross-Review + AI

Triad pairs • 45 minutes • Wrap

- 20 min: cross-review with 3 questions
- 10 min: AI critique
- 10 min: adopt / defer / reject
- 5 min: update Section 4 + provenance

⏱️ 20 + 10 + 10 + 5 · 15-min wrap

🎉 Day 4 Wrap

What we built

- Happy-path sequence (with latency)
- Sad-path sequence (with recovery)
- Weird-path sequence (with invariant)
- Cross-reviewed + AI-critiqued
- Technical Modeling Document (TMD) Section 4 drafted

What opens tomorrow

- **Performance baselines + monitoring**
- **Validating AI-generated data summaries**
- Friday integration: full TMD ships

Bring to Day 5: TMD Sections 1–4. Friday adds Section 5 and assembles the full TMD.

💬 Questions & Reflection

Which arrow on your happy path would break first under real load?